

MILLBROOK 3 W, NY, USA

WMO: 743690

Lat: 41.7857N

Lon: 73.7422W

Elev: 413

StdP: 14.48

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 64756

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.7	6.3	-10.5	3.2	3.0	-5.2	4.2	7.9	8.9	36.5	8.0	32.4	2.3	N/A	0.411

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.4	88.4	73.1	85.9	71.5	83.3	70.2	75.6	84.5	74.1	82.0	72.6	79.8	3.3	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.8	123.5	80.8	71.4	117.7	78.9	70.1	112.5	77.0	39.3	84.7	37.8	82.2	36.4	80.0	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.4	6.6	5.8	-7.9	93.6	6.0	2.4	-12.2	95.3	-15.7	96.7	-19.1	98.1	-23.5	99.8		
			-9.9	78.6	6.2	1.9	-14.4	80.0	-18.0	81.1	-21.5	82.2	-25.9	83.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.3	26.2	27.9	35.9	48.3	58.4	66.1	71.5	69.3	62.8	51.8	40.7	31.6
	DBStd	17.88	11.05	10.32	9.91	8.41	7.97	6.15	5.05	5.27	7.23	8.25	8.56	9.52
	HDD50	2904	739	620	452	129	14	0	0	1	77	299	573	
	HDD65	6281	1203	1038	903	507	238	59	8	18	128	415	729	1035
	CDD50	2661	1	2	14	77	275	483	667	599	385	135	20	3
	CDD65	561	0	0	0	5	34	91	210	152	62	7	0	0
	CDH74	4973	0	0	4	87	393	834	1852	1248	511	44	0	0
	CDH80	1382	0	0	0	24	98	214	584	327	130	5	0	0
Wind	WSAvg	2.6	2.9	3.1	3.2	3.4	2.6	2.1	1.9	1.9	1.9	2.5	2.7	2.7
Precipitation	PrecAvg	47.9	3.4	2.9	4.1	3.7	3.7	4.3	4.5	4.8	4.5	4.8	3.7	4.1
	PrecMax	63.0	7.4	8.5	6.7	7.8	6.7	8.4	10.3	10.5	11.2	17.4	6.9	7.6
	PrecMin	32.8	1.4	1.3	0.9	1.1	1.2	0.7	1.4	2.4	0.9	0.8	1.1	1.1
	PrecStd	6.7	1.6	1.5	1.4	1.6	1.5	2.2	2.1	2.1	2.8	3.6	1.4	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.6	62.0	71.7	84.1	86.9	90.0	92.7	90.2	88.6	78.8	68.8	62.3
		MCWB	56.3	53.9	57.2	61.1	69.4	74.4	75.7	77.3	73.7	66.9	59.0	58.8
	2%	DB	52.9	54.6	63.5	75.1	82.8	85.7	88.8	86.5	84.0	73.8	63.7	55.3
		MCWB	49.7	47.3	52.6	59.8	67.7	71.2	73.4	73.0	70.4	64.5	57.4	51.3
	5%	DB	45.9	48.1	56.4	69.7	78.6	82.1	86.0	83.6	79.8	70.0	59.9	50.3
		MCWB	41.8	41.5	47.2	56.0	65.0	68.3	72.1	70.9	68.5	62.6	52.6	46.5
	10%	DB	41.6	42.3	51.4	64.7	74.0	78.9	83.2	80.9	76.2	66.6	55.8	45.8
		MCWB	37.3	36.9	44.1	53.4	62.5	66.5	71.0	69.4	67.0	59.9	49.2	41.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.3	57.5	60.2	65.4	73.3	76.1	78.0	77.3	75.5	70.0	62.0	58.8
		MCDB	58.3	63.1	71.6	72.1	82.9	86.4	89.4	86.3	84.4	74.9	67.0	61.3
	2%	WB	48.6	51.4	55.5	62.0	70.6	72.9	76.0	75.0	73.1	67.3	58.5	52.8
		MCDB	50.5	55.4	64.2	70.4	80.1	81.2	85.6	82.5	80.3	71.1	62.6	54.9
	5%	WB	42.2	43.3	49.9	58.4	68.0	70.9	74.5	72.9	71.1	64.4	54.0	48.1
		MCDB	45.3	50.7	56.5	65.2	76.3	78.1	83.0	79.4	77.2	69.6	58.5	51.1
	10%	WB	37.9	39.2	45.0	54.2	65.7	68.7	73.1	71.2	69.1	60.9	49.5	43.9
		MCDB	41.0	43.8	51.0	61.8	72.3	75.3	80.7	77.6	74.5	65.4	54.8	46.6
Mean Daily Temperature Range	5% DB	MDBR	16.6	18.0	19.3	22.9	22.3	21.3	21.4	21.2	22.0	20.4	18.7	15.2
		MCDBR	20.4	23.9	27.5	32.1	27.6	25.2	24.4	23.3	24.9	24.4	23.8	19.0
		MCWBR	18.1	17.6	16.3	16.2	14.0	13.8	11.0	11.2	13.1	14.2	17.5	16.2
	5% WB	MCDBR	20.2	23.3	24.7	22.7	22.5	22.0	20.1	19.1	20.3	19.1	21.4	18.0
		MCWBR	18.6	18.2	16.6	14.9	12.5	12.8	10.3	10.2	11.6	13.5	17.8	17.1
Clear-Sky Solar Irradiance	taub	0.300	0.318	0.341	0.364	0.405	0.461	0.474	0.452	0.378	0.350	0.319	0.299	
	taud	2.400	2.366	2.395	2.383	2.295	2.173	2.147	2.200	2.384	2.472	2.513	2.493	
	Ebn at Noon	269	281	287	287	276	260	255	257	271	267	262	261	
	Edh at Noon	25	31	34	36	41	47	47	43	34	28	23	22	
All-Sky Solar Radiation	RadAvg	554	833	1155	1468	1651	1794	1861	1639	1347	888	614	451	
	RadStd	42	80	92	158	164	164	110	100	127	93	57	48	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (63 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+189	+99

Nomenclature: See separate page